机器学习

第 6 章 支持向量机

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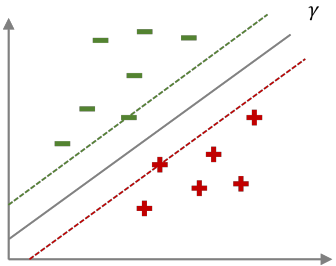
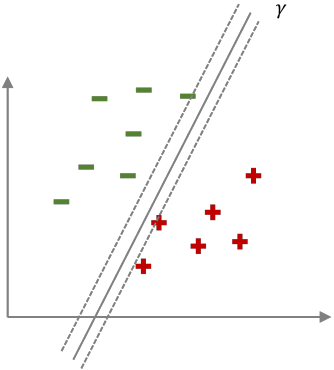
1. 线性支持向量机：间隔

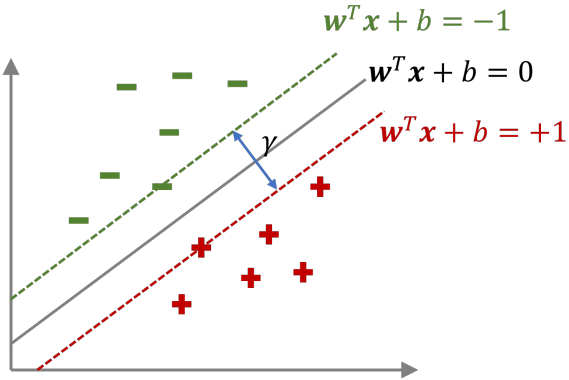
$$f(\boldsymbol{x}) = \text{sign}(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + b) = \begin{cases} 1, \boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + b \geq 0 \\ -1, \boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + b < 0 \end{cases}$$





训练支持向量机就是调整参数 w 和 b 使得间隔最大化

1. 线性支持向量机：间隔



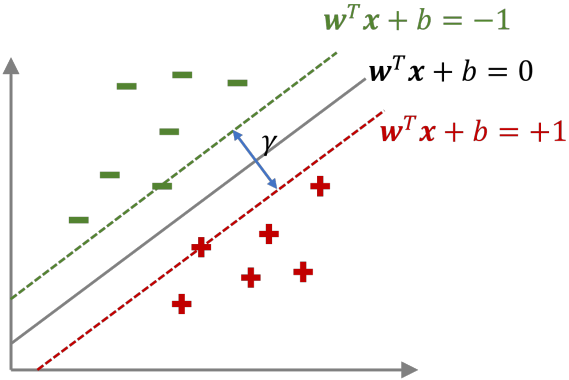
点坐标为 $\boldsymbol{y}$，直线方程为 $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}+b=0$，点到直线距离公式

$$d=\frac{|\boldsymbol{w}^{\mathrm{T}}\boldsymbol{y}+b|}{\|\boldsymbol{w}\|}$$

在直线 $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}+b=-1$ 取一个点 $\boldsymbol{y}$ 带入直线方程 $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}+b-1=0$ 中得

$$\gamma=\frac{|\boldsymbol{w}^{\mathrm{T}}\boldsymbol{y}+b-1|}{\|\boldsymbol{w}\|}=\frac{2}{\|\boldsymbol{w}\|}$$

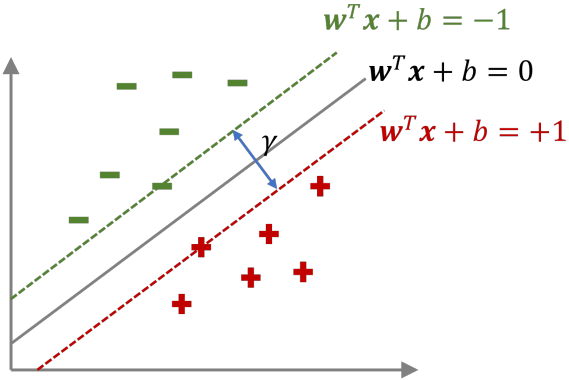
1. 线性支持向量机：间隔



优化问题

$$
\begin{aligned}
\text{maximize}: \quad & \gamma = \frac{2}{\|\boldsymbol{w}\|} \\
\text{subject to}: \quad & \boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b \geq +1, y_i = +1 \\
& \boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b \leq -1, y_i = -1 \\
& i = 1, 2, \cdots, m
\end{aligned}
$$

1. 线性支持向量机：间隔



优化问题

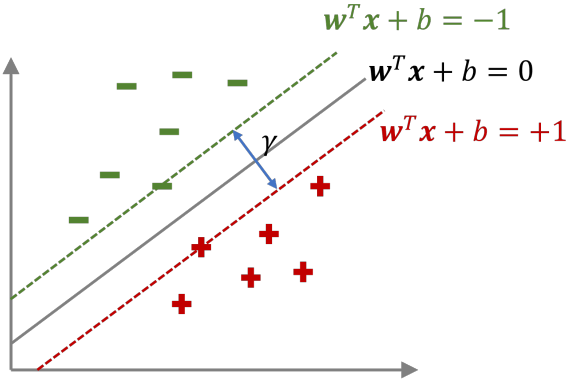
$$\text{minimize}: \quad \frac{1}{2}\|\boldsymbol{w}\|^2$$

$$\text{subject to}: \quad \boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b \geq +1, y_i = +1$$

$$\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b \leq -1, y_i = -1$$

$$i = 1, 2, \cdots, m$$

1. 线性支持向量机：间隔



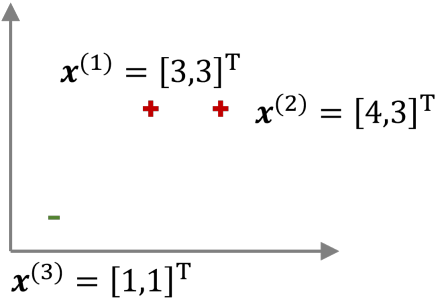
优化问题

$$\begin{aligned} \text{minimize}: \quad & \frac{1}{2}\|\boldsymbol{w}\|^2 \\ \text{subject to}: \quad & y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right) \geq +1 \\ & i = 1, 2, \cdots, m \end{aligned}$$

这个优化问题是**凸优化问题**

1. 线性支持向量机：作业 6

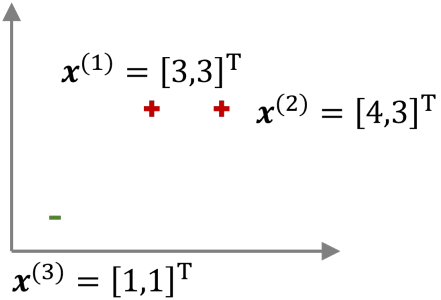
训练集共三个样本，其中正例为 $[3,3]^{\mathrm{T}}$，$[4,3]^{\mathrm{T}}$，负例为 $[1,1]^{\mathrm{T}}$。分类模型选用线性支持向量机，表达式为 $f = w_1x_1 + w_2x_2 + b$，写出支持向量机的原始优化问题。



$$
\begin{aligned}
\text{minimize}: \quad & \frac{1}{2}\|\boldsymbol{w}\|^2 \\
\text{subject to}: \quad & y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right) \geq +1 \\
& i = 1, 2, \cdots, m
\end{aligned}
$$

$$
\begin{aligned}
\text{minimize}: \quad & \frac{1}{2}\left(w_1^2 + w_2^2\right) \\
\text{subject to}: \quad & 3w_1 + 3w_2 + b \geq +1 \\
& 4w_1 + 3w_2 + b \geq +1 \\
& -\left(w_1 + w_2 + b\right) \geq +1
\end{aligned}
$$

1. 线性支持向量机：优化问题求解

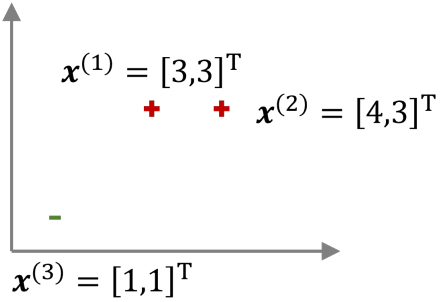


$$
\begin{aligned}
\text{minimize}: \quad & \frac{1}{2}\left(w_1^2 + w_2^2\right) \\
\text{subject to}: \quad & 3w_1 + 3w_2 + b \geq +1 \\
& 4w_1 + 3w_2 + b \geq +1 \\
& -\left(w_1 + w_2 + b\right) \geq +1
\end{aligned}
$$

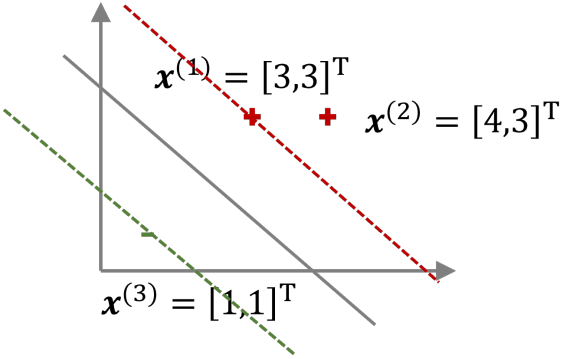
求约束优化问题可得：

$$
\begin{aligned}
& w_1 = 0.5, w_2 = 0.5, b = -2 \\
& f = \operatorname{sign}\left(0.5x_1 + 0.5x_2 - 2\right)
\end{aligned}
$$

1. 线性支持向量机：优化问题求解



$$f = \mathrm{sign}\,(0.5x_1 + 0.5x_2 - 2)$$



2. 对偶问题

原约束优化问题

$$
\begin{aligned}
\text{minimize}: \quad & \frac{1}{2}\|\boldsymbol{w}\|^2 \\
\text{subject to}: \quad & y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right) \geq +1 \\
& i = 1, 2, \cdots, m
\end{aligned}
$$

约束写成标准形式

$$
\begin{aligned}
\text{minimize}: \quad & \frac{1}{2}\|\boldsymbol{w}\|^2 \\
\text{subject to}: \quad & 1 - \left(y_i(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b)\right) \leq 0 \\
& i = 1, 2, \cdots, m
\end{aligned}
$$

2. 对偶问题

原约束优化问题

$$
\begin{aligned}
\text{minimize}: \quad & \frac{1}{2}\|\boldsymbol{w}\|^2 \\
\text{subject to}: \quad & 1-\left(y_i(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i+b)\right) \leq 0 \\
& i=1,2,\cdots,m
\end{aligned}
$$

拉格朗日乘子法

$$
\min_{\boldsymbol{w},b}\left\{\max_{\boldsymbol{\alpha}}\left[\frac{1}{2}\|\boldsymbol{w}\|^2+\sum_{i=1}^{m}\alpha_i\left(1-y_i(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i+b)\right)\right]\right\}
$$

拉格朗日函数

$$
L(\boldsymbol{w},b,\boldsymbol{\alpha})=\frac{1}{2}\|\boldsymbol{w}\|^2+\sum_{i=1}^{m}\alpha_i\left(1-y_i(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i+b)\right)
$$

2. 对偶问题

主问题（prime problem）

$$\min_{\boldsymbol{w},b}\left\{\max_{\boldsymbol{\alpha}}\left[\frac{1}{2}\|\boldsymbol{w}\|^2+\sum_{i=1}^{m}\alpha_i\left(1-y_i(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i+b)\right)\right]\right\}$$

对偶问题（dual problem）

$$\max_{\boldsymbol{\alpha}}\left\{\min_{\boldsymbol{w},b}\left[\frac{1}{2}\|\boldsymbol{w}\|^2+\sum_{i=1}^{m}\alpha_i\left(1-y_i(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i+b)\right)\right]\right\}$$

2. 对偶问题

对偶问题（dual problem）

$$\max_{\boldsymbol{\alpha}} \left\{ \min_{\boldsymbol{w},b} \left[\frac{1}{2}\|\boldsymbol{w}\|^2 + \sum_{i=1}^{m} \alpha_i \left(1 - y_i(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b)\right) \right] \right\}$$

导数与极值的关系

$$\frac{\partial L}{\partial \boldsymbol{w}} = \boldsymbol{w} - \sum_{i=1}^{m} \alpha_i y_i \boldsymbol{x}_i = \boldsymbol{0} \Rightarrow \boldsymbol{w} = \sum_{i=1}^{m} \alpha_i y_i \boldsymbol{x}_i$$

$$\frac{\partial L}{\partial b} = \sum_{i=1}^{m} \alpha_i y_i = 0$$

带入到原方程得

$$\max_{\boldsymbol{\alpha}} \left\{ \frac{1}{2}\|\sum_{i=1}^{m} \alpha_i y_i \boldsymbol{x}_i\|^2 + \sum_{i=1}^{m} \left[\alpha_i - \alpha_i y_i \left(\sum_{j=1}^{m} \alpha_j y_j \boldsymbol{x}_j \right)^{\mathrm{T}} \boldsymbol{x}_i - \alpha_i y_i b \right] \right\}$$

2. 对偶问题

$$\max_{\boldsymbol{\alpha}} \left\{ \frac{1}{2} \| \sum_{i=1}^{m} \alpha_i y_i \boldsymbol{x}_i \|^2 + \sum_{i=1}^{m} \left[\alpha_i - \alpha_i y_i \left(\sum_{j=1}^{m} \alpha_j y_j \boldsymbol{x}_j \right)^{\mathrm{T}} \boldsymbol{x}_i - \alpha_i y_i b \right] \right\}$$

进一步整理得

$$\max_{\boldsymbol{\alpha}} \left\{ \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j + \sum_{i=1}^{m} \alpha_i - \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j \right\}$$

合并同类项得

$$\max_{\boldsymbol{\alpha}} \left\{ \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j \right\}$$

2. 对偶问题

原约束优化问题

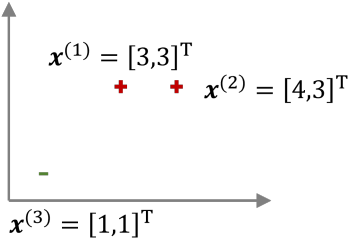
$$
\begin{aligned}
\text{minimize :} \quad & \frac{1}{2}\|\boldsymbol{w}\|^2 \\
\text{subject to :} \quad & y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right) \geq +1 \\
& i = 1, 2, \cdots, m
\end{aligned}
$$

对偶约束优化问题

$$
\begin{aligned}
\text{maximize :} \quad & \sum_{i=1}^{m} \alpha_i - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j \\
\text{subject to :} \quad & \sum_{i=1}^{m} \alpha_i y_i = 0 \\
& \alpha_i \geq 0 \\
& i = 1, 2, \cdots, m
\end{aligned}
$$

2. 对偶问题：作业 6

训练集共三个样本，其中正例为 $[3,3]^{\mathrm{T}}$，$[4,3]^{\mathrm{T}}$，负例为 $[1,1]^{\mathrm{T}}$。分类模型选用线性支持向量机，表达式为 $f = w_1x_1 + w_2x_2 + b$，写出支持向量机的对偶优化问题。



$$\text{maximize}: \quad \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j$$

$$\text{subject to}: \quad \sum_{i=1}^{m} \alpha_i y_i = 0$$

$$\alpha_i \geq 0$$

$$i = 1, 2, \cdots, m$$

$$\text{maximize:} \quad \sum_{i=1}^{3} \alpha_i - \frac{1}{2} \sum_{i=1}^{3} \sum_{j=1}^{3} \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j$$

$$= \alpha_1 + \alpha_2 + \alpha_3 - 9\alpha_1^2 - 21\alpha_1\alpha_2$$

$$+ 6\alpha_1\alpha_3 - 12.5\alpha_2^2 + 7\alpha_2\alpha_3 - \alpha_3^2$$

$$\text{subject to:} \quad \alpha_1 + \alpha_2 - \alpha_3 = 0$$

$$\alpha_i \geq 0, i = 1, 2, 3$$

2. 对偶问题：求解过程

$$
\begin{aligned}
\text{maximize:} \quad & \alpha_1+\alpha_2+\alpha_3-9\alpha_1^2-21\alpha_1\alpha_2+6\alpha_1\alpha_3-12.5\alpha_2^2+7\alpha_2\alpha_3-\alpha_3^2 \\
\text{subject to:} \quad & \alpha_1+\alpha_2-\alpha_3=0 \\
& \alpha_i \geq 0, i=1,2,3
\end{aligned}
$$

求解该线性约束二次规划问题得：$\alpha_1^\star=0.25, \alpha_2^\star=0, \alpha_3^\star=0.25$，$L(\boldsymbol{\alpha}^\star)=0.25$。根据 $\boldsymbol{w}=\sum_{i=1}^m \alpha_i y_i \boldsymbol{x}_i$ 得，

$$
\boldsymbol{w}^\star=\alpha_1^\star y_1 \boldsymbol{x}_1+\alpha_2^\star y_2 \boldsymbol{x}_2+\alpha_3^\star y_3 \boldsymbol{x}_3=0.25\begin{bmatrix}3\\3\end{bmatrix}+0\begin{bmatrix}4\\3\end{bmatrix}-0.25\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}0.5\\0.5\end{bmatrix}
$$

最优解满足龙格库塔（KKT）条件

$$
\alpha_i^\star\left[1-y_i\left(\boldsymbol{w}^{\star\mathrm{T}}\boldsymbol{x}_i+b^\star\right)\right]=0,\ i=1,2,\cdots,m
$$

带入 $\alpha_1^\star=0.25, y_1=1, \boldsymbol{w}^\star=[0.5,0.5]^{\mathrm{T}}, \boldsymbol{x}_1=[3,3]^{\mathrm{T}}$，解得 $b^\star=-2$。

2. 对偶问题：求解过程

原约束优化问题

$$
\begin{aligned}
\text{minimize}: \quad & \frac{1}{2}\left(w_1^2 + w_2^2\right) \\
\text{subject to}: \quad & 3w_1 + 3w_2 + b \geq +1 \\
& 4w_1 + 3w_2 + b \geq +1 \\
& -\left(w_1 + w_2 + b\right) \geq +1
\end{aligned}
$$

求解可得：

$$
\begin{aligned}
& w_1^\star = 0.5, w_2^\star = 0.5, b^\star = -2 \\
& f = \text{sign}\left(0.5x_1 + 0.5x_2 - 2\right)
\end{aligned}
$$

对偶约束优化问题

$$
\begin{aligned}
\text{maximize:} \quad & \alpha_1 + \alpha_2 + \alpha_3 - 9\alpha_1^2 - 21\alpha_1\alpha_2 \\
& + 6\alpha_1\alpha_3 - 12.5\alpha_2^2 + 7\alpha_2\alpha_3 - \alpha_3^2 \\
\text{subject to:} \quad & \alpha_1 + \alpha_2 - \alpha_3 = 0 \\
& \alpha_i \geq 0, i = 1, 2, 3
\end{aligned}
$$

求解可得：

$$
\begin{aligned}
& \alpha_1^\star = 0.25, \alpha_2^\star = 0, \alpha_3^\star = 0.25 \\
& w_1^\star = 0.5, w_2^\star = 0.5, b^\star = -2 \\
& f = \text{sign}\left(0.5x_1 + 0.5x_2 - 2\right)
\end{aligned}
$$

2. 对偶问题：求解过程

$$
\begin{aligned}
\text{maximize}: \quad & \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j \\
\text{subject to}: \quad & \sum_{i=1}^{m} \alpha_i y_i = 0 \\
& \alpha_i \geq 0,\ i = 1, 2, \cdots, m
\end{aligned}
$$

求出 $\boldsymbol{\alpha}^\star = [\alpha_1^\star, \alpha_2^\star, \cdots, \alpha_m^\star]$，$\boldsymbol{w}^\star = \sum_{i=1}^{m} \alpha_i^\star y_i \boldsymbol{x}_i$。

最优解满足龙格库塔（KKT）条件

$$
\alpha_i^\star \left[1 - y_i \left(\boldsymbol{w}^{\star\mathrm{T}} \boldsymbol{x}_i + b^\star\right)\right] = 0,\ i = 1, 2, \cdots, m
$$

任选一个支持向量 $\alpha_i^\star > 0$，根据 $1 - y_i \left(\boldsymbol{w}^{\star\mathrm{T}} \boldsymbol{x}_i + b^\star\right) = 0$ 求出 $b^\star$。

2. 对偶问题：支持向量与非支持向量

$$
\begin{aligned}
\text{maximize}: \quad & \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j \\
\text{subject to}: \quad & \sum_{i=1}^{m} \alpha_i y_i = 0 \\
& \alpha_i \geq 0,\ i = 1, 2, \cdots, m
\end{aligned}
$$

求出 $\boldsymbol{\alpha}^\star = [\alpha_1^\star, \alpha_2^\star, \cdots, \alpha_m^\star]$，再求出 $\boldsymbol{w}^\star$ 和 $b^\star$，最优解满足龙格库塔（KKT）条件

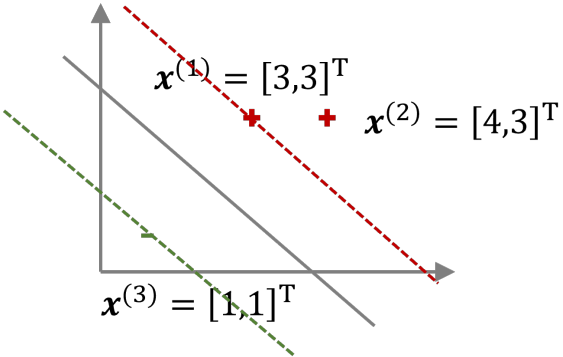
$$
\alpha_i^\star \left[1 - y_i \left(\boldsymbol{w}^{\star\mathrm{T}} \boldsymbol{x}_i + b^\star\right)\right] = 0,\ i = 1, 2, \cdots, m
$$

- $\alpha_i^\star > 0$，则 $1 - y_i \left(\boldsymbol{w}^{\star\mathrm{T}} \boldsymbol{x}_i + b^\star\right) = 0$，对应的样本在间隔线上，为支持向量。
- $\alpha_i^\star = 0$，则 $1 - y_i \left(\boldsymbol{w}^{\star\mathrm{T}} \boldsymbol{x}_i + b^\star\right) < 0$，对应的样本在间隔线外，为非支持向量。

2. 对偶问题：支持向量与非支持向量

- $\alpha_i^\star > 0$，则 $1 - y_i(\boldsymbol{w}^{\star\mathrm{T}}\boldsymbol{x}_i + b^\star) = 0$，对应的样本在间隔线上，为支持向量。
- $\alpha_i^\star = 0$，则 $1 - y_i(\boldsymbol{w}^{\star\mathrm{T}}\boldsymbol{x}_i + b^\star) < 0$，对应的样本在间隔线外，为非支持向量。

$$\alpha_1^\star = 0.25, \alpha_2^\star = 0, \alpha_3^\star = 0.25$$



2. 对偶问题：支持向量与非支持向量

$$
\begin{aligned}
\text{maximize}: \quad & \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j \\
\text{subject to}: \quad & \sum_{i=1}^{m} \alpha_i y_i = 0 \\
& \alpha_i \geq 0, \ i = 1, 2, \cdots, m
\end{aligned}
$$

支持向量机分隔超平面

$$
f(\boldsymbol{x}) = \operatorname{sign}\left(\boldsymbol{w}^{\star\mathrm{T}} \boldsymbol{x} + b^{\star}\right) = \operatorname{sign}\left(\sum_{i=1}^{m} \alpha_i^{\star} y_i \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x} + b^{\star}\right)
$$

- 决策函数关于 $\boldsymbol{x}$ 是线性函数。
- 只有支持向量（$\alpha_i > 0$）对最后决策有影响，非支持向量（$\alpha_i = 0$）没有影响。

2. 对偶问题：为什么引入对偶优化问题？

原约束优化问题

$$
\begin{aligned}
\text{minimize}: \quad & \frac{1}{2}\left(w_1^2 + w_2^2\right) \\
\text{subject to}: \quad & 3w_1 + 3w_2 + b \geq +1 \\
& 4w_1 + 3w_2 + b \geq +1 \\
& -\left(w_1 + w_2 + b\right) \geq +1
\end{aligned}
$$

求解可得：

$$
\begin{aligned}
& w_1^\star = 0.5, w_2^\star = 0.5, b^\star = -2 \\
& f = \text{sign}\left(0.5x_1 + 0.5x_2 - 2\right)
\end{aligned}
$$

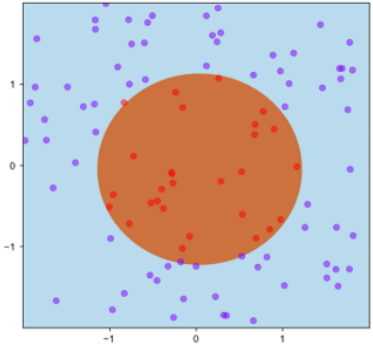
对偶约束优化问题

$$
\begin{aligned}
\text{maximize:} \quad & \alpha_1 + \alpha_2 + \alpha_3 - 9\alpha_1^2 - 21\alpha_1\alpha_2 \\
& + 6\alpha_1\alpha_3 - 12.5\alpha_2^2 + 7\alpha_2\alpha_3 - \alpha_3^2 \\
\text{subject to:} \quad & \alpha_1 + \alpha_2 - \alpha_3 = 0 \\
& \alpha_i \geq 0, i = 1, 2, 3
\end{aligned}
$$

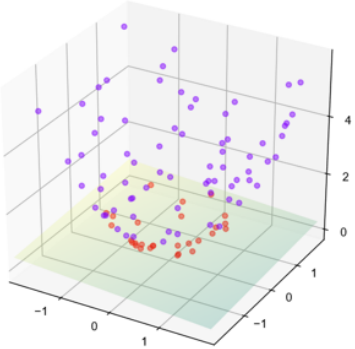
求解可得：

$$
\begin{aligned}
& \alpha_1^\star = 0.25, \alpha_2^\star = 0, \alpha_3^\star = 0.25 \\
& w_1^\star = 0.5, w_2^\star = 0.5, b^\star = -2 \\
& f = \text{sign}\left(0.5x_1 + 0.5x_2 - 2\right)
\end{aligned}
$$

3. 非线性支持向量机：映射空间



$$\phi(x_1, x_2) = [x_1, x_2, x_1^2 + x_2^2]$$



原始空间是有限维，一定存在一个高维空间使样本线性可分。

3. 非线性支持向量机：映射空间

$$\sum_{i=1}^{m}\alpha_i - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \boldsymbol{x}_i^{\mathsf{T}}\boldsymbol{x}_j \qquad f(\boldsymbol{x}) = \sum_{i=1}^{m}\alpha_i^{\star} y_i \boldsymbol{x}_i^{\mathsf{T}}\boldsymbol{x} + b^{\star}$$

映射函数 $\phi(\boldsymbol{x})$

$$\sum_{i=1}^{m}\alpha_i - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \phi(\boldsymbol{x}_i)^{\mathsf{T}}\phi(\boldsymbol{x}_j) \qquad f(\boldsymbol{x}) = \sum_{i=1}^{m}\alpha_i^{\star} y_i \phi(\boldsymbol{x}_i)^{\mathsf{T}}\phi(\boldsymbol{x}) + b^{\star}$$

用核函数替代向量内积

$$\sum_{i=1}^{m}\alpha_i - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) \qquad f(\boldsymbol{x}) = \sum_{i=1}^{m}\alpha_i^{\star} y_i \kappa(\boldsymbol{x}_i, \boldsymbol{x}) + b^{\star}$$

3. 非线性支持向量机：核函数

$$\sum_{i=1}^{m}\alpha_i - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j)$$

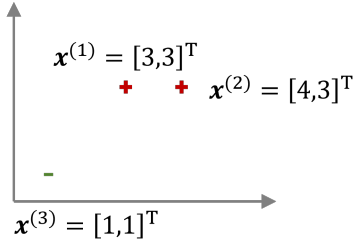
$$f(\boldsymbol{x}) = \sum_{i=1}^{m}\alpha_i^\star y_i \kappa(\boldsymbol{x}_i, \boldsymbol{x}) + b^\star$$

- 线性核 $\kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) = \boldsymbol{x}_i^{\mathrm{T}}\boldsymbol{x}_j$
- 多项式核 $\kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) = (\boldsymbol{x}_i^{\mathrm{T}}\boldsymbol{x}_j)^d$
- 高斯核 $\kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) = \exp\left(-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2}{2\sigma^2}\right)$
- 拉普拉斯核 $\kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) = \exp\left(-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|}{\sigma}\right)$
- sigmoid 核 $\kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) = \tanh(\beta\boldsymbol{x}_i^{\mathrm{T}}\boldsymbol{x}_j + \theta)$

1. 一个核函数对应一个映射函数，不同参数的核函数对应不同映射函数。
2. 把找映射函数的问题转化为找核函数的问题。

3. 非线性支持向量机：示例

训练集共三个样本，其中正例为 $[3,3]^{\mathrm{T}}$，$[4,3]^{\mathrm{T}}$，负例为 $[1,1]^{\mathrm{T}}$。分类模型选用非线性支持向量机，核函数选择高斯核 $\kappa(\boldsymbol{x}_i,\boldsymbol{x}_j)=\exp\left(-\frac{\|\boldsymbol{x}_i-\boldsymbol{x}_j\|^2}{2\sigma^2}\right),\sigma=1$。写出非支持向量机的对偶优化问题。



$$\text{maximize}: \quad \sum_{i=1}^{m}\alpha_i-\frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \kappa(\boldsymbol{x}_i,\boldsymbol{x}_j)$$

$$\text{subject to}: \quad \sum_{i=1}^{m}\alpha_i y_i=0$$

$$\alpha_i\geq 0,\ i=1,2,\cdots,m$$

$$\text{maximize:} \quad \sum_{i=1}^{3}\alpha_i-\frac{1}{2}\sum_{i=1}^{3}\sum_{j=1}^{3}\alpha_i\alpha_j y_i y_j \kappa(\boldsymbol{x}_i,\boldsymbol{x}_j)$$

$$=\alpha_1+\alpha_2+\alpha_3-0.5\alpha_1^2-0.5\alpha_2^2-0.5\alpha_3^2$$

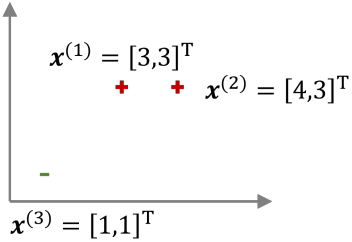
$$-\exp(-0.5)\alpha_1\alpha_2+\exp(-4)\alpha_1\alpha_3+\exp(-6.5)\alpha_2\alpha_3$$

$$\text{subject to:} \quad \alpha_1+\alpha_2-\alpha_3=0$$

$$\alpha_i\geq 0, i=1,2,3$$

3. 非线性支持向量机：示例

训练集共三个样本，其中正例为 $[3,3]^{\mathrm{T}}$，$[4,3]^{\mathrm{T}}$，负例为 $[1,1]^{\mathrm{T}}$。分类模型选用非线性支持向量机，核函数选择高斯核 $\kappa(\boldsymbol{x}_i,\boldsymbol{x}_j)=\exp\left(-\frac{\|\boldsymbol{x}_i-\boldsymbol{x}_j\|^2}{2\sigma^2}\right),\sigma=1$。写出非支持向量机的对偶优化问题。



$$\text{maximize}: \quad \sum_{i=1}^{m}\alpha_i-\frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \kappa(\boldsymbol{x}_i,\boldsymbol{x}_j)$$

$$\text{subject to}: \quad \sum_{i=1}^{m}\alpha_i y_i=0$$

$$\alpha_i\geq 0,\ i=1,2,\cdots,m$$

$$\text{maximize:} \quad \sum_{i=1}^{3}\alpha_i-\frac{1}{2}\sum_{i=1}^{3}\sum_{j=1}^{3}\alpha_i\alpha_j y_i y_j \kappa(\boldsymbol{x}_i,\boldsymbol{x}_j)$$

$$=\alpha_1+\alpha_2+\alpha_3-0.5\alpha_1^2-0.5\alpha_2^2-0.5\alpha_3^2$$

$$-0.6065\alpha_1\alpha_2+0.0183\alpha_1\alpha_3+0.0015\alpha_2\alpha_3$$

$$\text{subject to:} \quad \alpha_1+\alpha_2-\alpha_3=0$$

$$\alpha_i\geq 0, i=1,2,3$$

3. 非线性支持向量机：求解过程

$$\begin{aligned}
\text{maximize}: \quad & \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) \\
\text{subject to}: \quad & \sum_{i=1}^{m} \alpha_i y_i = 0 \\
& \alpha_i \geq 0,\ i = 1, 2, \cdots, m
\end{aligned}$$

求出 $\boldsymbol{\alpha}^\star = [\alpha_1^\star, \alpha_2^\star, \cdots, \alpha_m^\star]$，求出 $\boldsymbol{w}^\star$。最优解满足龙格库塔条件求出 $b^\star$

$$\alpha_i^\star \left[1 - y_i \left(\boldsymbol{w}^{\star\mathrm{T}} \boldsymbol{x}_i + b^\star\right)\right] = 0,\ i = 1, 2, \cdots, m$$

得到非线性支持向量机决策函数：

$$f(\boldsymbol{x}) = \operatorname{sign}\left(\sum_{i=1}^{m} \alpha_i^\star y_i \kappa(\boldsymbol{x}_i, \boldsymbol{x}) + b^\star\right)$$

3. 非线性支持向量机：求解过程

$$\text{maximize}: \quad \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j)$$

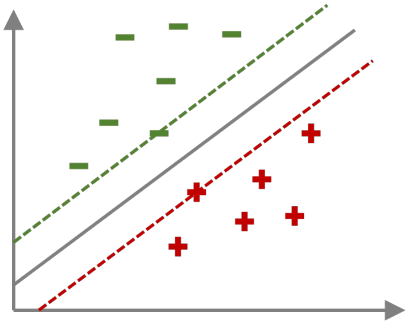
$$\text{subject to}: \quad \sum_{i=1}^{m} \alpha_i y_i = 0, \ \alpha_i \geq 0, \ i = 1, 2, \cdots, m$$

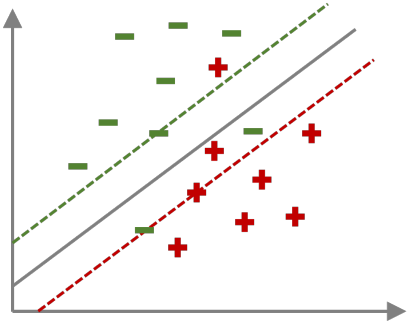
得到非线性支持向量机决策函数：

$$f(\boldsymbol{x}) = \text{sign}\left(\sum_{i=1}^{m} \alpha_i^\star y_i \kappa(\boldsymbol{x}_i, \boldsymbol{x}) + b^\star \right)$$

- 决策面关于 $\boldsymbol{x}$ 是非线性的。
- 只有支持向量（$\alpha_i > 0$）对决策有影响，非支持向量（$\alpha_i = 0$）没有影响。

4. 软间隔支持向量机





$$\begin{aligned} \text{minimize}: \quad & \frac{1}{2}\|\boldsymbol{w}\|^2 \\ \text{subject to}: \quad & y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right) \geq 1 \\ & i = 1, 2, \cdots, m \end{aligned}$$

$$\begin{aligned} \text{minimize}: \quad & \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{m}\xi_i \\ \text{subject to}: \quad & y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right) \geq 1 - \xi_i \\ & \xi_i \geq 0 \\ & i = 1, 2, \cdots, m \end{aligned}$$

4. 软间隔支持向量机：对偶问题

$$
\begin{aligned}
\text{minimize}: \quad & \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{m}\xi_i \\
\text{subject to}: \quad & y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right) \geq 1 - \xi_i \\
& \xi_i \geq 0 \\
& i = 1, 2, \cdots, m
\end{aligned}
$$

不等式写成标准形式

$$
\begin{aligned}
\text{minimize}: \quad & \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{m}\xi_i \\
\text{subject to}: \quad & 1 - \xi_i - y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right) \leq 0 \\
& -\xi_i \leq 0 \\
& i = 1, 2, \cdots, m
\end{aligned}
$$

4. 软间隔支持向量机：对偶问题

$$
\begin{aligned}
\text{minimize}: \quad & \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{m}\xi_i \\
\text{subject to}: \quad & 1-\xi_i - y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right) \le 0 \\
& -\xi_i \le 0 \\
& i = 1, 2, \cdots, m
\end{aligned}
$$

拉格朗日函数

$$
L(\boldsymbol{w}, b, \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{\mu}) = \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{m}\xi_i + \sum_{i=1}^{m}\alpha_i\left[1-\xi_i - y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right)\right] + \sum_{i=1}^{m}\mu_i(-\xi_i)
$$

主问题

$$
\min_{\boldsymbol{w}, b, \boldsymbol{\xi}} \left\{ \max_{\boldsymbol{\alpha}, \boldsymbol{\mu}} \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{m}\xi_i + \sum_{i=1}^{m}\alpha_i\left[1-\xi_i - y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right)\right] + \sum_{i=1}^{m}\mu_i(-\xi_i) \right\}
$$

4. 软间隔支持向量机：对偶问题

主问题

$$\min_{\boldsymbol{w},b,\boldsymbol{\xi}} \left\{ \max_{\boldsymbol{\alpha},\boldsymbol{\mu}} \ \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{m}\xi_i + \sum_{i=1}^{m}\alpha_i\left[1-\xi_i - y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right)\right] + \sum_{i=1}^{m}\mu_i(-\xi_i) \right\}$$

对偶问题

$$\max_{\boldsymbol{\alpha},\boldsymbol{\mu}} \left\{ \min_{\boldsymbol{w},b,\boldsymbol{\xi}} \ \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{m}\xi_i + \sum_{i=1}^{m}\alpha_i\left[1-\xi_i - y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right)\right] + \sum_{i=1}^{m}\mu_i(-\xi_i) \right\}$$

4. 软间隔支持向量机：对偶问题

对偶问题

$$\max_{\boldsymbol{\alpha},\boldsymbol{\mu}} \left\{ \min_{\boldsymbol{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{m}\xi_i + \sum_{i=1}^{m}\alpha_i\left[1-\xi_i - y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b\right)\right] - \sum_{i=1}^{m}\mu_i\xi_i \right\}$$

$$\frac{\partial L}{\partial \boldsymbol{w}} = \boldsymbol{w} - \sum_{i=1}^{m}\alpha_i y_i \boldsymbol{x}_i = \boldsymbol{0} \Rightarrow \boldsymbol{w} = \sum_{i=1}^{m}\alpha_i y_i \boldsymbol{x}_i$$

$$\frac{\partial L}{\partial b} = \sum_{i=1}^{m}\alpha_i y_i = 0$$

$$\frac{\partial L}{\partial \boldsymbol{\xi}} = \boldsymbol{C} - \boldsymbol{\alpha} - \boldsymbol{\mu} = \boldsymbol{0} \Rightarrow \alpha_i + \mu_i = C$$

4. 软间隔支持向量机：对偶问题

$$\max_{\boldsymbol{\alpha},\boldsymbol{\mu}} \left\{ \min_{\boldsymbol{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{m}\xi_i + \sum_{i=1}^{m}\alpha_i\left[1-\xi_i-y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i+b\right)\right] - \sum_{i=1}^{m}\mu_i\xi_i \right\}$$

将表达式带入目标函数

$$\begin{aligned}\max_{\boldsymbol{\alpha},\boldsymbol{\mu}} \quad & \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}}\boldsymbol{x}_j + C\sum_{i=1}^{m}\xi_i + \sum_{i=1}^{m}\alpha_i - \sum_{i=1}^{m}\alpha_i\xi_i - \sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}}\boldsymbol{x}_j \\ & - \sum_{i=1}^{m}\alpha_i y_i b - \sum_{i=1}^{m}\mu_i\xi_i\end{aligned}$$

合并同类项，整理得

$$\max_{\boldsymbol{\alpha}} \left\{ \sum_{i=1}^{m}\alpha_i - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}}\boldsymbol{x}_j \right\}$$

4. 软间隔支持向量机：对偶问题

硬间隔支持向量机

$$\max: \quad \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j$$

$$\text{s.t.}: \quad \sum_{i=1}^{m} \alpha_i y_i = 0$$

$$\alpha_i \geq 0,\ i = 1, 2, \cdots, m$$

龙格库塔条件

$$\alpha_i^\star \left[1 - y_i \left(\boldsymbol{w}^{\star\mathrm{T}} \boldsymbol{x}_i + b^\star\right)\right] = 0,\ i = 1, 2, \cdots, m$$

软间隔支持向量机

$$\max: \quad \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{x}_j$$

$$\text{s.t.}: \quad \sum_{i=1}^{m} \alpha_i y_i = 0$$

$$0 \leq \alpha_i \leq C,\ i = 1, 2, \cdots, m$$

龙格库塔条件

$$\alpha_i^\star \left[1 - \xi_i^\star - y_i \left(\boldsymbol{w}^{\star\mathrm{T}} \boldsymbol{x}_i + b^\star\right)\right] = 0$$

$$\mu_i^\star \xi_i^\star = 0, i = 1, 2, \cdots, m$$